DI22007

Design Project #2

Group 8

Mechanical Clock

Design Project

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Abstract Introduction	Design Concept Design Implementation	Assembly and Testing	Discussion Conclusion Future Work

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Content

1. Abstract 3

2. Introduction 3

3. Design Concept 4

3.1. Basic areas 4

3.2. Gear train 5

3.3. Materials chosen 7

4. Design Implementation 7

4.1. Design software 7

4.2. Gear design 8

4.3. Time regulation design 9

4.4. Power source design 10

4.5. Fabrication method 11

5. Assembly and Testing 12

5.1. Assembly 12

5.2. Adjustment 14

5.3. Testing 14

6. Discussion 15

7. Future work 16

8. Conclusion 16

9. References 17

1. Abstract

The project aims to construct a mechanical clock with mostly acrylic material. The main idea of the work is to design and calculate the quantities of the teeth of the gears. The number of teeth makes sure the ratios between the gears meet the requirements of the absolute ratios between the hour hand, the minute hand and the second hand, which is 1/60 and 1/12 respectively. The design also needs to achieve the less shaft as possible. After designing and calculating, the initial assembly is created on SolidWorks to prove whether the clock could work properly, and the side view of each component was exported as DXF file for the final sketching on AutoCAD. Once the sketch was completed, a given acrylic board will be cut as the sketch drew. The final product will be assembled and tested whether the clock could function in the right way.

2. Introduction

The overall process contains designing, sketching, and testing. The designing process mostly includes the quantities of teeth of gears and the different combinations of gears. With the solid ratios between three kinds of hands, there are various types of ways to design gears, which also leads to the initial attempts of laying examples of the combinations and listing them as a table as shown on the Fig. 1.

Gear Ratio	Ratio Department Design									
60:1	2:1	3:1	2:1	2:1	2:1	6:1	5:1	4:1	3:1	2:1
	2:1	4:1	5:1	3:1	2:1	10:1	12:1	15:1	20:4	30:1
	3:1	5:1	6:1	10:1	15:1					
	5:1									

Min gear teeth	12

Ratio	Gear 1	Gear 2
3:1	17	51
4:1	56	14
5:1	60	12

Fig. 1 The table of the gear combinations ([redacted])

When all the combinations are listed, these combinations are sketched on the paper to calculate and simulate how many shafts are required and whether the shafts could be combined in the right way. For example, if there are three shafts, the connection between the shafts' centres must meet the shape of the triangle, which means the sums of three shafts' number of teeth need to follow the rule of the triangular inequality. It is because the number of teeth represents the radius of the gear once its module is determined. The condition of four shafts follows the same rule, which requires the lines connected by four shafts could be combined as a quadrilateral. Finally, a combination of 4 shafts is chosen as the ultimate design.

After the selection of the gear combination, it comes to the decision of the power source. As the category of the clock is limited to the purely mechanical clock, the range of the power sources is two: gravity masses and mainspring. With the given material is mostly acrylic and the parts are mostly created by laser-cutter on an acrylic board, it is impossible to use the mainspring as the power source, which leads to the choice of gravity masses. To use the gravity masses as the power source, it is not as simple as hanging a subject on a shaft. It is because the gravity masses cannot secure the accuracy of time, which brings the thought of the time regulator.

The time regulator makes sure the clock's time interval remain constant. The designed clock is a mechanical clock, which needs a mechanical regulator such as escapement. This section uses the distances and angles of different components to limit the activity area, which also limit the period of regulator with each oscillation. The period of regulator is calculated as:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

'L' represents the length of the pendulum and 'g' represents the gravity.

The time representation of the designed clock uses the hour hand, the minute hand and the second hand. The hour hand has 12 scales, while the minute hand and the second hand have 60 scales. And the materials used in the whole process are acrylic and PLA. Most components such as gears and panels are made of the acrylic board by laser-cutter. The escape wheel of the time regulator is made by 3D-printer because its concavities and extrusion cannot be formed by simple laser-cutter.

3. Design Concept

3.1. Basic areas

1. Power source:

The designed clock chooses gravity masses as the power source. The reason is that the clock must be a purely mechanical clock, which makes the decision of the power source come out from two options: gravity masses and mainspring. However, the main components of the clock are created from an acrylic board. And the mainspring produced by 3D-printer will be hard to analyse and be combined with the gears with insufficient calculation and simulation. Thus, the design picks gravity masses as the power source, which only takes the length of the pendulum into consideration.

2. Time regulation:

In the design of the time regulation, Graham Clock Escapement and weight-driven power source are chosen. It is effective to reach the main goal of precise time regulation, simple installation and fabrication by above design concepts.

3. Time display:

The display of time selects three different hands as the representations. The three hands are the hour hand, the minute hand and the second hand respectively. The reason behind the choice is that the designed mechanical clock has three specific gears, which rotate at $\frac{\pi}{30} rad/s, \frac{\pi}{1800} rad/s$ and $\frac{\pi}{21600} rad/s$. (Fig. 2) Three angular speeds are in line with the rates of the second, minute and hour. Once the three hands are attached to those gears with locked rotation and maintain the same speeds with them, the mechanical clock could be created and present time precisely.

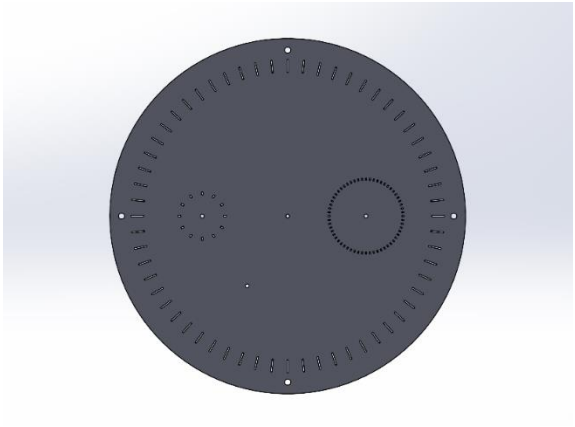


Fig. 2 The front view of the dial ([redacted])

3.2. Gear train

An intricately designed gear train is employed in the project, meticulously configured to ensure precise timekeeping. The heart of the design is a well-thought-out selection of gear ratios and stages, realizing the movement transition between the time regulation system, the energy source and the time display.

Considering the standard temporal relationship between these units of time, we opted for a gear ratio of 60:1 to harmoniously translate the rotation of the second hand to the subtle of the minute hand, and 12:1 for minute hand to hour hand. This critical reduction is achieved through a strategic multi-stage gear train configuration. Rather than using a singular ratio, a compound approach is a better way to enhance efficiency and accuracy.

First viable implementation of this strategy is to divide the 60:1 reduction into three distinct stages: an initial 3:1 gear pair followed by a subsequent 4:1 ratio, continuing to be followed by a 5:1 gear pair. (Fig. 3) This configuration not only maintains the necessary overall

reduction, but also contributes to smoother operation and reduced backlash, enhancing the watch's reliability and longevity.

In the same way, another arrangement for minute to hour could incorporate two 2:1 ratio in the first two stage, paired with a subsequent 3:1 ratio, (Fig. 4) achieving the same aggregate reduction while potentially offering benefits in terms of gear size, depending on the specific constraints and design objectives of the clock.

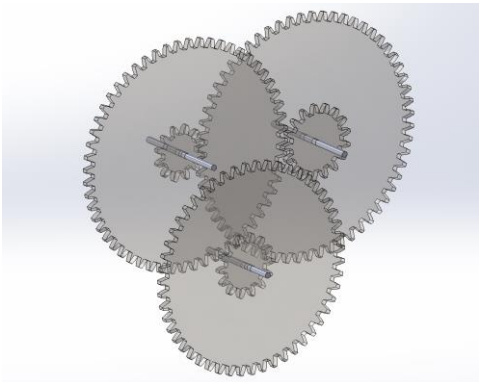


Fig. 3 Distinct stages for 60:1 ([redacted])

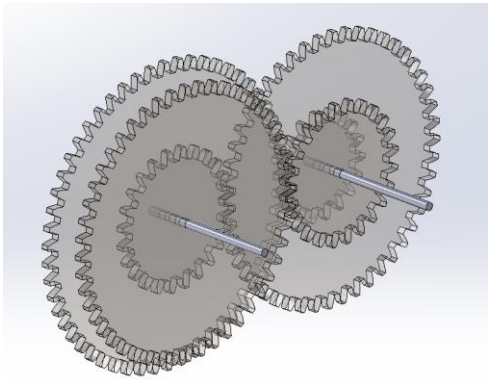


Fig. 4 Distinct stages for 12:1 ([redacted])

However, when the difference of stages between two gears is odd, their direction of rotation will be reversed. (Fig 5) Obviously, the three-stage design between the second hand and the minute hand cannot satisfy the requirement that both need rotate clockwise. Thus, another gear ratio 1:1 is added to transform the direction of the rotation from second hand to minute hand. (Fig. 6) As the difference between the second hand and hour hand is 6, rather an odd, they rotate in the same direction.

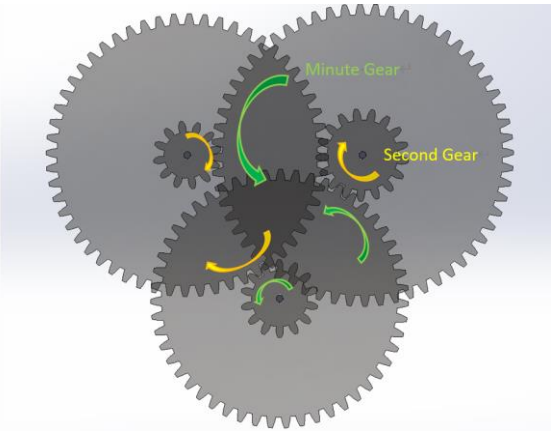


Fig. 5 Rotation case in distinct stages for 60:1 ([redacted])

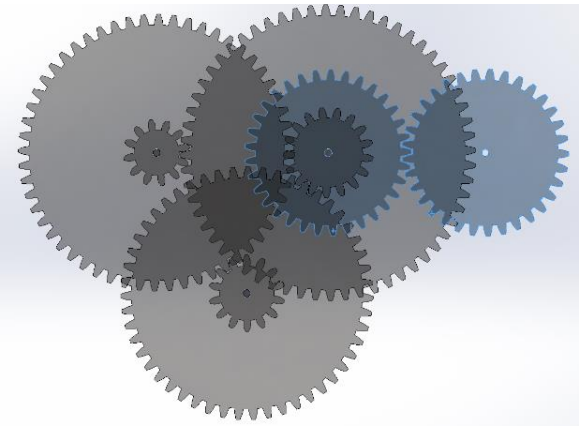


Fig. 6 Add for 1:1 stage ([redacted])

The choice of these gear ratios and staging is underpinned by a thorough analysis of fabrication efficiency, power consumption, and the desire for a balanced distribution of wear across the gear train components. Although the design of multi-stage gears will increase the difficulty in the manufacturing process and prolong the manufacturing time, this is the approach which can minimize the gear ratio. A lower gear ratio brings a less torque amplified from the driven wheel. As torque decreases, the contact pressure between the gears may

drop, especially on the smaller gears, which often need to transmit large driving forces. This decreased pressure can lead to the avoidance that increased friction on the contact surfaces of the gears exacerbate wear and frictional losses.

3.3. Materials chosen

The materials used for the construction of the clock are two: acrylic and PLA. The acrylic is chosen for its high transparency, high workability, and high ductility. The high transparency eases the design difficulty because the issue of the covers between gears and panels can be removed. High workability and ductility make the given acrylic board easy to be processed by the laser-cutter. PLA is mostly applied to the construction of the time regulator. The time regulator contains the pallets, the escape wheel, and the pendulum. The only part needs to be 3D-printed is the escape wheel. The 3D-printer could provide supports so that the produced escape wheel will be safer and less risk. It is because that the escape wheel is not a simple wheel with the unchanging thickness, which contains an extra part that is impossible to be created on an acrylic board as shown on the figure 7.

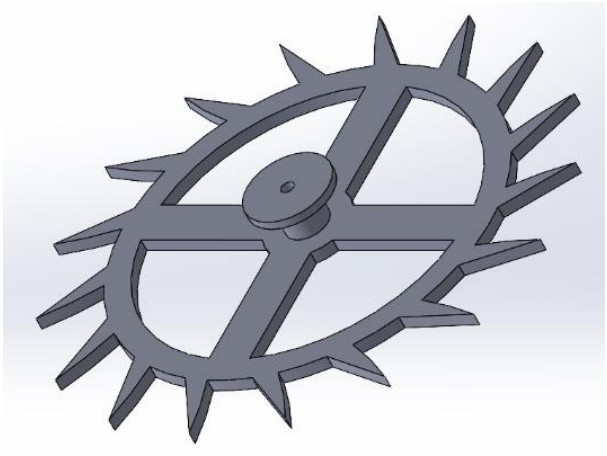


Fig. 7 The escape wheel ([redacted])

4. Design Implementation

4.1. Design software

In the process of designing the acrylic mechanical clock, state-of-the-art software tools were applied for facilitated modelling, intricate design adjustments, including AutoCAD and SolidWorks, both of which played vital roles in transforming conceptual sketches into tangible, manufacturable designs.

AutoCAD, flagship product from Autodesk, is the benchmark for computer-aided design software, empowering professionals in architecture, engineering, and manufacturing. This

multi-function tool facilitates the efficient creation and editing of 2D designs, enabling designers to build complex geometries on the screen with unmatched flexibility. (Ahmad, 2022)

Besides the basic work done in AutoCAD, as the main platform for 3D modelling and motion simulation, SolidWorks was most frequently used in the designing part. Its prowess in solid modelling enabled the team to construct precise, realistic 3D representations of each part and the assembly as a whole. Allowing for quick modifications and the exploration of multiple design render SolidWorks becoming non-displaceable. As well, SolidWorks provides a visual testing platform for altered motion status over time, possible to foresee if the gears rotate properly and if the product go wrong after a period. (*SOLIDWORKS Simulation*, 2023)

4.2. Gear design

The gears chosen for the designed clock are all spur gears on SolidWorks as shown in Fig. 8, which is because the design clock does not require any angle or groove to make components connect to each other more efficiently. The plane of spur gear should be flat to make sure the power is transmitted gently with the connection method of gears is adhesion. The curvature will only make the assembly harder to operate. Once the shape of the gears is determined as the spur gear, the number of teeth of the gears could be implemented and make a draft.

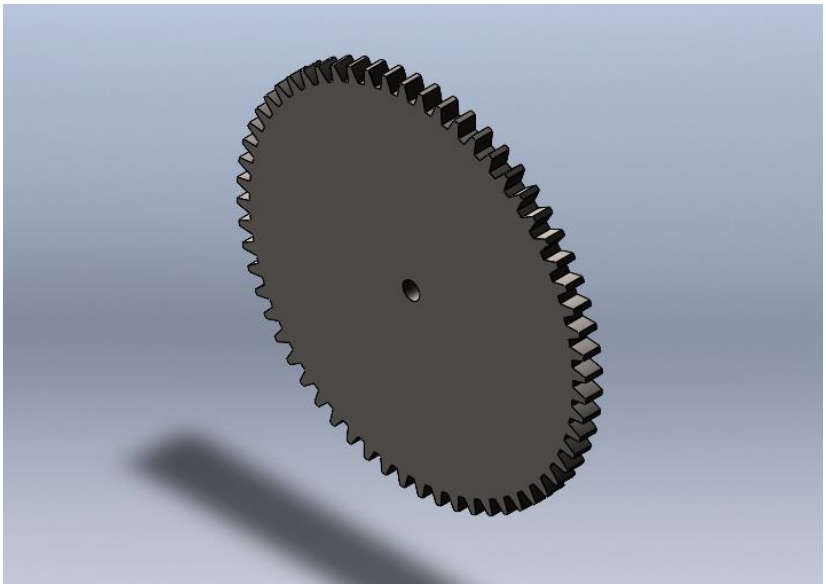


Fig. 8 Then spur gear model in SolidWork ([redacted]*)*

The design of the support panels is based on the design of the clock. The designed clock is a circular clock, which leads to the idea of round panels. The front view of the whole clock is as shown in figure 9. The dial contains 8 holes besides the scales, which are set for the connections between two panels through shafts. However, two panels are insufficient to hold the whole clock without vertical support. Thus, two rectangular shaped holes are created on the back side panel as shown in Fig. 9, which lifts the final product.

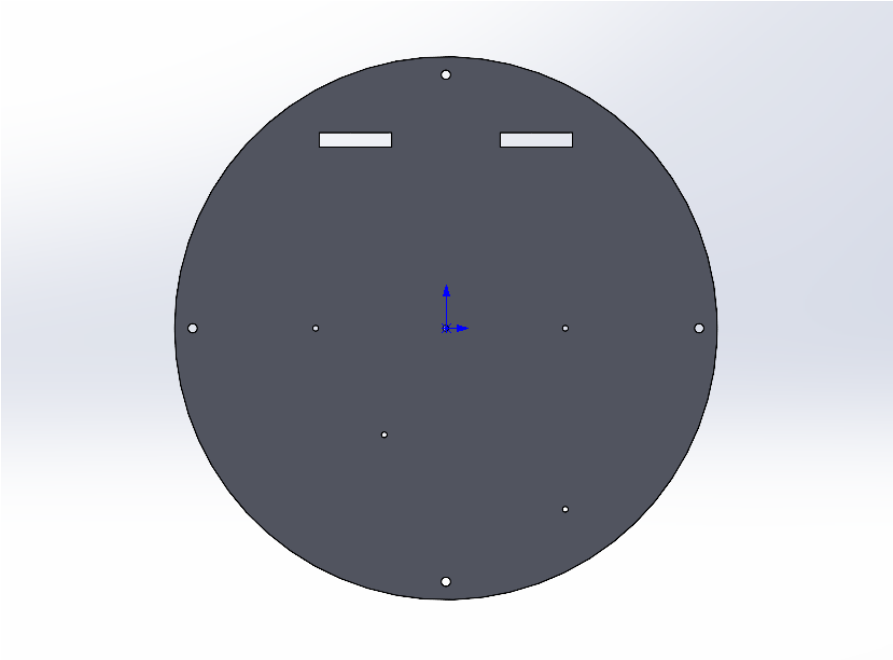


Fig. 9 The backside panel ([redacted])

The gears have different numbers of teeth, which meet the ratios between the hour hand and the minute hand (1/12), and the minute hand and the second hand (1/60). It is the only index which is different among gears. The pitch diameter and the pressure angle stay the same during the construction of the clock, which are 3.1416 and 14.5 degrees. It is because the module of the gears is 2, the pitch is equal to the product of the module and π.

$$p = Pi \times Module = \pi m.$$

Thus, the pitch diameter is limited. The indexes are determined as same for the smooth rotation between gears. The gear can rotate gently with same pressure angle to avoid the over or resistive transmission. The same module makes sure the radii of gears are proportional to their numbers of teeth, which helps to find the accurate positions of the centres of gears. The gears will function as designed when each gear is located at the proper place.

4.3. Time regulation design

a) Reason of choosing Graham Clock Escapement

The Graham Clock Escapement is chosen which is also known as the deadbeat escapement was chosen in the project. The integral device has three main components: Escape Wheel, Pawl and power equipment. By using this system, the escape wheel only moves when the pendulum passes the central to avoid recoil as a result develop the accuracy and improve its durability.

Also compared to the Detent Escapement, the clock is ease of manufacture. The components are relatively easy to assemble and manufacture. Also, less frequent maintenance is required with less wear between escape wheel and pawl.

b) Calculation

As for the design of Pendulum Length, the length of the pendulum is the main influence factor for time regulation. As the formula shows:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

As the acceleration due to gravity g is approximately $9.8m/s^2$.

To make the clock with a 1.2second period and swings a half-0.6second each way the pendulum length can be calculated:

$$L = (\frac{T}{2\pi})^2 g = (\frac{1.2}{2\pi})^2 \times 9.8 \approx 357.5mm$$

As for mass and shape, to make sure constant motion choose a Disk-like shape bob at the end of the pendulum to minimize the air resistance. Although mass doesn't affect the time regulation ideally, it would be better to choose a proper mass of pendulum which can keep motion steady and regular to overcome friction during each component in practical experiments.

c) Other choices for time regulation

Except from pendulum, balance wheel and hairspring can also be another choice for the time regulation of escapement mechanism.

Balance wheel oscillates regulated by the hairspring. While the moment of inertia I and the stiffness K which is also known as spring constant of the hairspring determine the oscillation period mainly. From above oscillation frequency f can be calculated as:

$$f = \frac{1}{2\pi}\sqrt{\frac{K}{I}}$$

Subsequent Regulation is done by altering the stiffness and effective length of the hairspring. Additionally, timing screws is also used to change the moment of inertia for adjusting oscillation frequency then.

4.4. Power source design

a) Reason of choosing Weight-Driven power source

Weight-driven clock uses gravitational force as its power source. Gravitational force is generated by suspended weight through a cord or chain, driving the clock's gear train. The main advantage of Weight-driven is consistent power supply by weight which ensure operation stably and accurate time regulation. Also, Low consumption and simple structure are obvious strengths.

While, there also has some drawbacks. For example, weight driven needs sufficient place for descending weights. Then, it is necessary to load heavy again manually after one time release which is troublesome in daily life.

Compared to other way of power source such as hairspring, Weight-driven produces more stable force rather than a force that gradually decreases as the spring is released.

b) Method of synchronizing the escapement with the power source

The driven gravity is produced by the weight which relates to escape wheel by cord. Descending weight drives escape wheel. Then, pawl of the escapement regulates the release of energy to ensure escape wheel influence the second hand turns accurately in sync with the pendulum's oscillations.

c) Calculation

It is necessary to determine the Required Torque which is depended on the friction with each of components and motion of pendulum. while in ideal condition, friction is always negligible.

To determine the Weight and Force, ensure the Torque T_e at the Escape Wheel and T_d at the Driving Drum (Wikipedia contributors, 2024):

$$T_d = T_e \times Total\ Gear\ Ratio$$

Then calculate the Required Force F:

$$F = T_d / r$$

Finally, weight W is got converted from force:

$$W = F / g$$

During the actual experiment, calculation should be verified and adjusted according to the actual situation. For example, adapt weight when clock is faster or slower than expected calculations during the experimental process.

4.5. Fabrication method

a) Fabrication method

A combination of laser-cutter and 3-D printer is chosen in the fabrication. As for most of main components, such as standard part gears and dial fit laser-cutter which are accurate and exact. Also, the pawl in the escapement system must be cut by laser because its length is not able

to be printed on the 3-D printer with its insufficient space. While the escapement wheel should be printed by 3-D printer in the reason of hollow in the center which is designed to hanging the heavy is inconvenient to cut acrylic by laser.

b) Affection on design approach

The combination of the two-fabrication method can maximize the advantages of each of them, while there also has some affection on design cause of different fabrication method choice. Laser cutting makes it possible to cut standard parts such as gears, shafts and dials that need to be finely machined accurately as designed without large errors. However, where laser cutting is inconvenient, such as in the case of escapements, it is easier to use 3D printing to replace the grooves and overhangs. In addition, in the actual installation, it is not difficult to find, before cutting the design of the draft stage needs to be reserved for the size of the cavity, such as some distance, to prevent the inability to snap in due to errors and other issues.

5. Assembly and Testing

5.1. Assembly

After discussion and adjustment, the clock obtained a transmission scheme with eight sets of meshing gears. Using a ratchet as the driving gear, gear rotation times of one minute and one hour were realized through appropriate proportional transmission.

The transmission types between the sixteen gears were mainly divided into two types: meshing transmission and coaxial transmission. Meshing transmission was achieved through two gears with the same modulus and type, while coaxial transmission was achieved by adding a connecting shaft between the gears that need to rotate at the same angular speed to bond the two gears together. And the adhesion between the connecting shaft and the coaxial rotating gear is carried out through glue.

After the meshing and coaxial connection of the gear set were bonded, the layout of the gears was connected and fixed using a long shaft, which was also the most important step in assembly. According to the previous gear layout design, a total of four long shafts were used to connect the gears, with three shafts connected to three pointers. Because only three gears, which control the three pointers speed, were directly bonded to the corresponding long shaft, and all other gears needed to generate relative rotation with the long shaft. To realize the relative motion more smoothly, bearings were used between gears and long shafts, including the front and rear clock dial. (Fig. 10)



Fig. 10 The bearing in the clock dial ([redacted])

During the assembly process, it was particularly important to determine the position and spacing of gears. According to the design before, there was a distance of three millimeters between each gear face. Considering the need to maintain gear spacing to ensure gear engagement in the same plane, nuts were used to control the distance and instead of the connecting shaft being cut (Fig. 11). And the determination of position often leads to errors, so margin needs to be left when modeling.

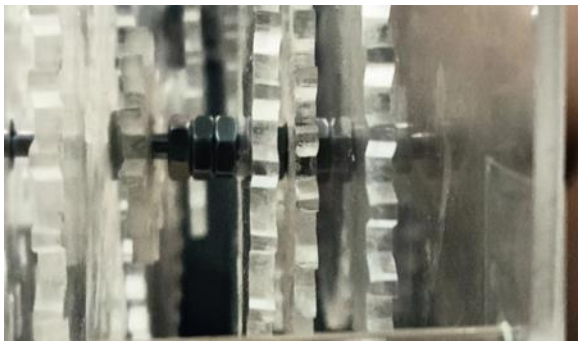


Fig. 11 The nuts connection between gears ([redacted])

Then, it was the most difficult part— Graham escapement. When modeling and assembling, the relative position between the pawl and the center of the escapement wheel calculated before the experiment allows the pawl to coordinate and freely swing with the escapement wheel. To allow the pawl to rotate freely, bearings were also used at the connection between the long shaft and the pawl. Then, by adjusting the position of the bearing on the shaft, the claw and wheel were in the same plane to achieve their function

After the long shaft and gears were matched and arranged, the final step was to fix the long shaft to the front and rear dial plates. Because the shafts connected by the three pointers also needed to move relatively to the dial, bearings were added between the three shafts and the front and rear dial. After using bearings to fix and connect the long shafts to the panels, apply Nanogel to the outside of the area where the panel and bearings fit to prevent the front and rear movement of the long shaft from affecting the gear meshing

Finally, fix the pointers to the corresponding long shaft and obtain a complete and beautiful clock.

5.2. Adjustment

During the assembly process, adjustments were made to some details for convenience and aesthetics.

The first adjustment was about the connection of gears rotating at the same angular speed. When using glue to bond the nut with the front and rear gears, due to the small contact area, the bonding area is prone to loosening, resulting in the failure of the connecting shaft function. Therefore, the cut connecting shaft is used between the outer rings of the gears, close to the connecting shaft, to make the connection between the gears tighter and enable to withstand the forces and moments generated by transmission meshing. (Fig. 12)

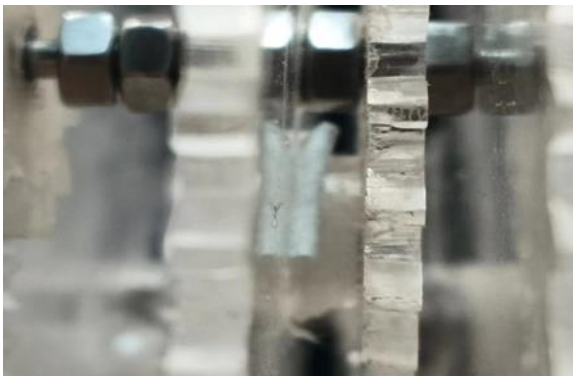


Fig. 12 Fixed outer ring of connecting shaft ([redacted]*)*

The second adjustment is about the relative motion and adhesion issues between some gears and long shafts. In the design, only the gears that controlled the rotation of the pointer was considered to be bonded to the long shafts, but it was not considered that the gears that rotates at the same angular speed as the controlling gears also rotated together with the long shaft. So due to the issues of distance and fixed rotation, the gears that rotated with the shaft were completely fixed to the long shafts with glue.

5.3. Testing

After finishing the assembly and adjustment, the clock was hung on the wall for testing. And a thick rope was wrapped around the protruding shaft of the escapement wheel to provide power.

The test results did not meet expectations. The gear set can achieve normal meshing and rotation, but the pawl of the escapement mechanism could not control the rotation cycle time of the escapement wheel due to its thin lateral thickness, which cannot achieve the effect of timing. Therefore, ultimately, the rotation and time transfer of the three pointers could only achieve by manually rotating the driving wheel.

6. Discussion

The clock functions through a unique mechanism where a rope, wrapped tightly around the Escapement Wheel, is pulled down by a heavy weight. This descending force causes the escapement wheel to rotate. However, to ensure a controlled and precise movement, an escaper, known as a Pawl, engages with the teeth of the escapement wheel. This fork acts as a regulator, limiting the rotation of the wheel to a specific tooth pitch in a cycle of precisely 1.2 seconds. Once the escapement wheel rotates in this manner, it triggers a series of gear interactions. The teeth of these gears mesh with each other, driving other gears to rotate, ultimately powering the clock's movement. The source of this energy is the gradual release of the gravitational potential energy of the falling weight, which is converted into kinetic energy to keep the clock ticking.

In the actual test, to ensure that the gear can rotate smoothly, we use the hand to apply force instead of the weight to apply gravity as Fig. 13 shown below.



Fig. 13 The test of the clock ([redacted])

The clock's design incorporates three separate axes for the second, minute, and hour hands. The longest hand, positioned in the middle, represents the seconds and its scale spans the entire dial. The second-longest hand, on the left, indicates the minutes, while the shortest hand on the right shows the hours. Each of these hands has its own dedicated scale for easy reading. Consequently, before setting the clock, the user must determine the current time by reading from the centre outwards, first to the left for minutes and then to the right for hours.

However, a significant challenge arises due to the narrow design of the escapement fork. This narrowness fails to adequately limit the rotation of the escapement wheel, resulting in an irregular and non-periodic movement. This malfunction prevents the clock from running smoothly and accurately, making it impossible to calibrate with precise time measurements.

Even if this issue with the escapement fork is addressed, the clock's reliance on a heavy weight as its power source remains a limitation. The length of the rope determines how many

rotations the escapement wheel can complete, which in turn limits the clock's runtime. Unfortunately, relying solely on the weight of the rope is insufficient to power the clock for more than an hour, making it difficult to verify the accuracy of its operation over a prolonged period. This shortcoming necessitates further improvements to the clock's design and power mechanism.

7. Future work

During the process of designing manufacturing assembly and operation, the members found some problems about the design that could be improved in the future.

The first is the manufacturing problem. During the manufacturing process, team members found that the coaxial design made assembly more difficult and time-consuming. Too many gears are designed, that is, there are too many gears on the axis controlling the second hand, the minute hand, and the hour hand, so that the error caused by the bite of each pair of gears leads to the final gear operation. There is also a large error expected at the time of design compared with the actual gap, resulting in assembly is not smooth. For example, the aperture of the gear is the same as the diameter of the shaft, resulting in the shaft cannot be smoothly installed into the gear, and the distance between the gears is too small, resulting in interference between the non-occluding gears.

Then there are the problems encountered in the testing section. The design of the escapement caused a big problem, and the type mismatch between the escapement wheel and the escapement fork directly led to the failure of the clock operation. At the same time, the existence of multiple layers of gears in one shaft leads to the problem that multiple layers of gears do not mesh, which seriously affects the accuracy of the clock.

In addition to the future in the design can be improved, in the manufacturing also need to be improved. Errors need to be considered when purchasing manufacturing materials, for example, a shaft with a smaller aperture than the design should be purchased to facilitate assembly. At the same time, it is necessary to consider the use of more appropriate methods and tools for assembly, such as the use of nuts to control the distance between the gears will cause the gears to sway from side to side and affect the mesh,also improve the glue reinforcement when assembling the coaxial can speed up the assembly efficiency.

8. Conclusion

This project is about designing, assembling and testing a mechanical clock. The design of this clock mainly consists of three parts: dial, gear and escapement. The dial is the most common disk, where the hands are separately read. The part of gear design using multi-stage gear train to make the gears as concentrated as possible, reducing the use of rotating shafts. The escapement adopts Graham Clock Escapement and gravity action system, so that the overall time of the clock can be precisely controlled and limited.

Testing after assembly, the group found that the gears could turn as desired, but the clock failed due to an error in the escapement design that could not limit the escapement wheel. In conclusion, there are some problems in this design and manufacturing that lead to the failure of the final test, and more improvements in the future can realize the operation of the clock.

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